1. **What is a computer system?**

   A computer system is a combination of hardware and software that works together to process data and produce meaningful information. It performs input, processing, storage, and output operations.

2. **Why is a computer called a digital machine?**

   A computer is called a digital machine because it processes data in binary form (0s and 1s). All operations are based on digital signals.

3. **What is binary number system?**

   The binary number system uses only two digits: 0 and 1. It is the fundamental number system used by computers to represent data.

4. **What is a bit?**

   A bit (binary digit) is the smallest unit of data in a computer. It can have only two values: 0 or 1.

5. **What is ASCII and how many characters does it represent?**

   ASCII (American Standard Code for Information Interchange) is a coding system used to represent characters in computers. It represents 128 standard characters.

6. **What is logic 0 and logic 1?**

   Logic 0 represents a LOW signal (off/false), while logic 1 represents a HIGH signal (on/true) in digital systems.

7. **What is an analog signal?**

   An analog signal is a continuous signal that varies smoothly over time. It can have infinite values within a range.

8. **What is a digital signal?**

   A digital signal is a discrete signal that has only specific values, usually 0 and 1. It is used in digital electronics and computers.

9. **Give two differences between analog and digital signals.**

Analog Signals	**Digital Signals**

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Analog signals are continuous	Digital signals are discrete
Analog signals are more affected by noise	Digital signals are more reliable.

10. Which signal is more resistant to noise and why?

Digital signals are more resistant to noise because they use fixed values (0 and 1), making errors easier to detect and correct.

11. What is a logic gate?

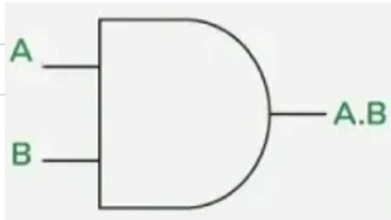
A logic gate is an electronic device that performs logical operations on one or more inputs to produce a single output.

12. Why are logic gates called building blocks of digital circuits?

Logic gates are called building blocks because all digital circuits are made by combining multiple logic gates to perform complex operations.

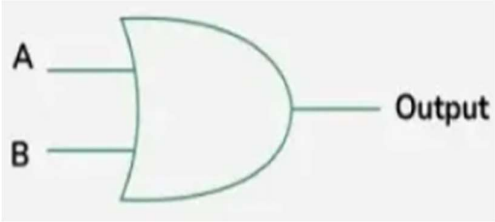
13. Define AND gate?

In and gate, when all inputs are 1 the output will be 1, else it will be 0. An AND gate have 2 or more than 2 inputs. Its logic expression for 2 input is F = A . B & for 3 it is F = A . B . C.



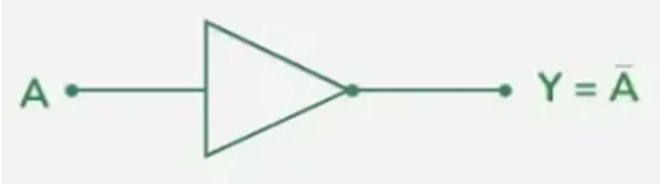
14. Define OR gate?

In or gate, if any input is 1 the output will be one, else it will be 0. It has 2 or more than 2 inputs. Its logic expression for 2 input is F = a + b & for 3 input F = a + b + c.



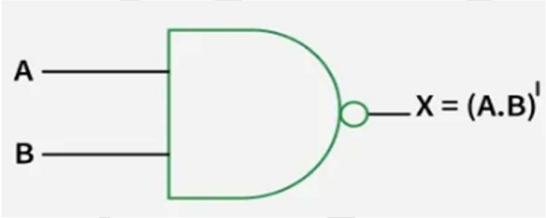
15.Define NOT gate?

NOT gate is a single input gate. If input is 1 then the output will be 0, if input is 0 the output will be 1. Its logic expression is F = X'.



16.Define NAND gate?

NAND gate is the combination of two gates NOT and AND gate. NAND gate converts the output of and gate. Its logic expression is F = (a . b)'.



17.Define NOR gate?

NOR gate combines two gates NOT and OR gate. NOR gate converts the output of OR gate. It logic expression is F = (A + B)'.

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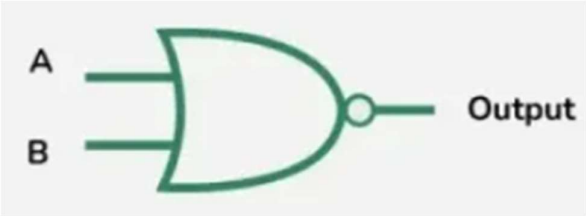
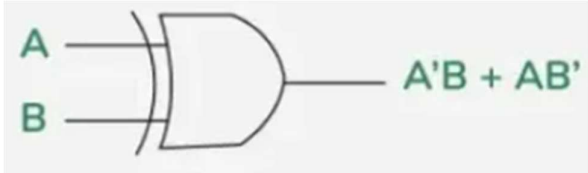


18.Define XOR gate?

Exclusive-OR gate has the same symbol as OR gate but with an extra curved at input. It produces output 1 only when there are different inputs,

if the inputs are same the output will be 0. Its logic expression is F = x'y + xy'.



19.Write the truth table of AND gate.

A	B	F= A.B
0	0	0
0	1	0
1	0	0
1	1	1

20.Write the truth table of OR gate.

A	B	F= A + B
0	0	0
0	1	1
1	0	1
1	1	1

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21.Write the truth table of NAND gate.

A	B	F= (A . B)'
0	0	1
0	1	1
1	0	1

1	1	0

22. Write the truth table of NOR gate.

A	B	F= (A + B)’
0	0	1
0	1	0
1	0	0
1	1	0

23. Write the truth table of NOT gate.

A	F= A’
1	0
0	1

24. Write the truth table of XOR gate.

A	B	F= A + B
0	0	0
0	1	1
1	0	1
1	1	0

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25. What is a Boolean function?

A Boolean function is a mathematical expression that uses binary variables and logical operations like AND, OR, and NOT.

26. What is Karnaugh Map (K-Map)?

A Karnaugh Map is a graphical method used to simplify Boolean expressions. It helps reduce the number of variables and logic gates.

27. Why is K-Map used for simplification?

K-Map is used to simplify Boolean expressions to reduce circuit complexity, cost, and number of logic gates used.

28. State De Morgan's Theorem.

De Morgan's Theorem states:

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

29. What are use of logic Gates?

They are used in **memory circuits** to **store binary data (0 and 1)**.

They are used in **control systems** to **control and manage signals**.

They are used in **data encoding and decoding** to **convert data for communication**.

They are used in **mathematical calculations** to **perform arithmetic operations**.

They are used in **digital signal processing** to **process and filter signals**.

They are used in **encryption and decryption** to **secure data**.

They are used in **calculator circuits** to **perform calculations quickly**.

They are used in **traffic light systems** to **control traffic flow**.

They are used in **robotics systems** to **make decisions and control movement**.

They are used in **security systems** to **control alarms and access**.

They are used in **automotive systems** to **control vehicle functions (ABS, airbags)**.

They are used in **home automation systems** to **automate devices (lights, temperature)**.

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They are used in **medical devices** to **monitor and control functions**.

They are used in **aerospace systems** to **control navigation and guidance**.

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